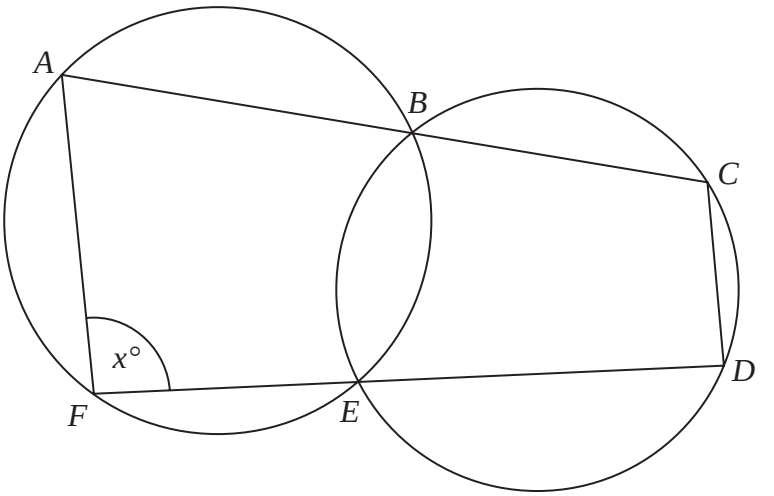
22



A, *B*, *E* and *F* lie on a circle.
B, *C*, *D* and *E* lie on a different circle.

ABC and *FED* are straight lines.
Angle *AFD* = *x*°

Prove that *AF* and *CD* are parallel.

Angle AFE = x° because F, E, D are collinear.

Angle ABE = (180 – x)°

1 mark
Opposite angles of cyclic quadrilateral ABEF total 180°.
Angle EBC = x°

1 mark
ABC is straight: angle ABE + angle EBC = 180°.
Angle CDE = (180 – x)°

1 mark
Opposite angles of cyclic quadrilateral BCDE total 180°.
Angle AFE + angle CDE = 180°.

These are co-interior angles on transversal FD.

Therefore AF is parallel to CD.

1 mark
Co-interior angles sum to 180°, so the two lines are parallel.

(Total for Question 22 is 4 marks)

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